

Noe Noah Studio — Phases 3 & 4 Research: Sketch-to-Color Configurator and Autonomous Collection Assistant

Bottom line: build Phase 3 so that color and lamination are computed, not generated: a deterministic "layer-stack" compositor or a PBR (Three.js/Blender) renderer driven by real supplier swatch data, plus a DFM rules engine. Use diffusion models only as an optional, masked "photographic finishing" pass whose output is checked against the swatch colors. No existing product combines eyewear-specific lamination modelling, real acetate stock, sketch-zone editing and DFM checks. The nearest competitors are VARAi (CAD→3D eyewear configurator with a supplier-grade material library), Vizcom (generic sketch-to-render with Color Match and Region Maps) and Mazzucchelli's own swatch configurator. Each covers one slice, which leaves a real gap for a creator-brand tool.

TL;DR

- **The market is fragmented, not empty.** Eyewear tools such as VARAi, Fittingbox, PlanetVA and Thema Optical's iGreen configurator render existing CAD or 3D frames in catalog colors. Generic AI studios such as Vizcom and Raspberry AI turn sketches into renders but know nothing about acetate laminations, supplier SKUs or hinge tenons. Nobody publicly offers "B&W technical sketch → designer-drawn lamination zones → render in real supplier stock → manufacturability check → sampling-ready variants."
- **Generative image models are the wrong source of truth for color and lamination edges.** Even 2025–2026 research aimed at exact hex/RGB control (NumColor, ColorWave, Paint-Anything) only shows progress on benchmarks, and lamination stripes are geometry. Compute color and edges deterministically from swatch data and layer thicknesses. Use diffusion (Qwen-Image-Edit, FLUX, SDXL+ControlNet) only for lighting and material realism inside locked masks, then measure and correct against swatch Lab values (ΔE_{2000}).
- **LLMs belong in orchestration, not pixels.** Use them for supplier-catalog extraction (with Docling/vision-LLM plus human validation), palette and lamination suggestion (constrained to in-stock SKUs and scored by trend and brand-DNA models), and agentic collection generation. DFM guardrails must be a deterministic geometry and rules engine. Recommended MVP: React + Konva/Paper.js editor, FastAPI + PostgreSQL/pgvector, Docling + VLM extraction, a 2.5D compositor + Three.js MeshPhysicalMaterial preview, optional ComfyUI/Qwen-Image-Edit finishing on serverless GPUs.

Key Findings

1. Prior art and competitors — how close they get to the spec

Eyewear-specific

Player	What it does	What it does NOT cover	Closeness to spec
VARAi 3D Configurator / Design Studio	Converts CAD to "1:1-accurate 3D" models, applies "a supplier-grade library of acetates, metals, lenses, and finishes... swapped in real time with physically based (PBR) rendering," and renders studio photography; it also says it can generate 3D models "from just a front and side image."	Starts from CAD or photos, not from editable B&W sketches; no evidence of user-drawn lamination zones, layer-stack edge rendering, DFM rules, or trend/brand-DNA generation.	Closest on Phase 3 rendering + real materials. Main competitor/partner candidate.
Mazzucchelli 1849 configurator ("Shape your ideas")	The supplier's own tool: "Try the different colors to see what the final result would be" with products sold as 7 × 17 cm Creative Cases; catalog organized by Havanas, Gradient, Multilayer, Laminated, Monocolor; product pages list thickness and MOQ (e.g., laminated 906303 "Encounter": 4.0 and 7.0 mm, 600 × 1400 mm, "MOQ 1 sheet /	Single-supplier, template-based, no sketch input, no DFM, no brand learning.	Key data source, and a signal that suppliers see value in visualization. Its 2026 catalog is published as an XLSX file , which is far easier to ingest than PDF.

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	1 strip"). It also publishes an Eyewear Trends 2027 trend book with Peclers Paris.		
Fittingbox	Virtual try-on leader; "195.000+ frames" in its database; "3D from Photo" builds basic 3D models from two photos; studio digitization in Toulouse/Miami takes about 3 weeks.	Post-design (retail) only; digitizes existing physical frames.	Adjacent (downstream: try-on of approved designs).
PlanetVA, Perfect Corp, Pic Copilot, GlassesUSA "Pairfect Match"	3D viewers/configurators, AR try-on, face-shape recommendation, AI model photos.	Consumer-facing; no design-side tooling.	Low.
Thema Optical iGreen Custom, Cubitts Bespoke, Topology	Consumer configurators with fixed shapes and many colors (iGreen lists "597 colors"; Cubitts "over 130 acetate colours"; Topology "50 designer colors" with "Acetate from Mazzuchelli, Italy").	Fixed catalog models; color swap on a known 3D model; no new design generation.	Proves the "configurator on real stock" UX and that per-part color selection works commercially.
Frame Maker (Andrés Roppa, ShapeDiver/Grasshopper)	Turns "2D outlines into ready-to-prototype 3D frames," automating "the mechanical or	Geometry-focused, 3D-printing oriented; no photoreal acetate/lamination rendering or AI suggestion.	Very relevant to the 2D→3D step if you go the PBR route.

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	repetitive parts of the modeling process."		
Academic: "Automated Custom Sunglasses Frame Design Using AI and Computational Design"	AI + Grasshopper parametric pipeline from facial scans to a personalized frame, with manufacturability validation.	Fit-driven, not aesthetic/material-driven.	Shows that parametric + AI eyewear pipelines are feasible.

Adjacent fields

Player	Relevance	Gap vs. spec
Vizcom (industrial/footwear design)	Sketch-to-render; Color Match ("Control if Vizcom stays true to the colors of your sketch"); Region Maps ("Segment any product render into individual parts, then edit color, material, and pattern on each one"); Custom Palettes/Styles trained on "4 to 30 images"; Extract Colors (capped at 10 colors); 2D→3D.	The closest generic analogue to Phase 3 + brand style. But it is general-purpose and has no supplier SKU binding, lamination physics or DFM. Its brand-accurate color claims are vendor marketing and not independently verified.
Raspberry AI (fashion)	Sketch-to-render, print generation, on-body; \$24M Series A led by Andreessen Horowitz (Jan 2025), about 10 months after a \$4.5M seed; TechCrunch reported 70 customers including Under Armour, Gruppo Teddy and MCM Worldwide.	Apparel; no material-stock binding or rigid-product DFM.
BLNG (jewelry)	Sketch/text-to-photoreal 3D jewelry renders; according to Crunchbase News it has	Jewelry, not acetate.

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	raised \$4.5 million, including a \$3 million seed round in April 2025.	
Heuritech (trend forecasting, acquired by Luxurysight on Dec 2, 2024)	Computer vision detects "2,000+ product attributes" in social images; claims "92% accuracy, 24 months ahead."	Apparel-centric; accuracy figures are self-reported. A possible data input for the Phase 4 trend radar, not a design tool.
Research: ZeST (ECCV 2024), MaTe (2026), MatSwap	Zero-shot material transfer from an exemplar image onto an object using IP-Adapter + inpainting + depth/shading cues.	Transfers the "look" of a swatch photo but does not guarantee exact color or layer edges. Useful for texture realism inside masks.
Research: NumColor, ColorWave, Color Mapper, Paint-Anything	Methods for hex/RGB-precise color in diffusion. NumColor reports a "10–30× improvement" in color-harmony error over baselines; ColorWave claims "exact RGB-level color control... without fine-tuning."	Research-grade, benchmarked on generic objects; NumColor itself documents failure modes. Not production guarantees for supplier references.

Differentiation opportunities (the gap you can own):

1. **Lamination as a first-class data object:** a layer stack with ordered SKUs, thicknesses and cut angles, shown visibly on bevels, temple edges and the inside of the frame front. Nobody models this publicly.
2. **Stock-bound configurator:** every color choice is a real SKU with thickness, sheet size, MOQ and availability, sourced from Mazzucchelli XLSX/PDF, Takiron, Jimei and other catalogs.
3. **Sketch-first input** (the industry's native 2D technical drawing) rather than CAD.
4. **DFM-in-the-loop:** hinge and tenon anchoring, minimum widths, groove depth and sheet-thickness checks before any render is approved.
5. **Brand DNA + trend scoring** that constrains generation to the house style, so outputs are not generic AI aesthetics.

2. The core risk: generative models vs. exact supplier colors and lamination edges

- **Why diffusion fails here.** Text encoders such as CLIP "are not designed to interpret raw numerical data like RGB or Hex values". Models "fail to interpret numerical colors such as hex codes" (NumColor). Diffusion also re-lights and re-tones the whole image, so even an exact input color drifts after shading, gloss and white balance. Lamination edges are a geometric consequence of layer thickness $\times$ bevel angle $\times$ milling path; a diffusion model will make up plausible-looking stripes that do not match the real stack (wrong order, wrong proportions, stripes that appear where the cut would not expose them).
- **Translucency compounds the problem.** Many acetates are transparent, milky or tortoise. Perceived color depends on thickness and background, so a single hex value is not even the right descriptor. You need swatch images (front-lit and back-lit), plus measured Lab values where available.
- **Hybrid solution (recommended).**
 1. **Deterministic base layer:** render each zone with a texture taken from the actual supplier swatch image (tiled or synthesized). Compute lamination edges procedurally: from the layer stack (e.g., pink 2 mm / red 2 mm / cream 3 mm) and the bevel geometry, draw the exposed stripes on side views, temple tips and the inner front.
 2. **Physically based preview:** extrude the 2D vector outline into a 2.5D/3D mesh and use Three.js `MeshPhysicalMaterial`, which supports `transmission`, `thickness`, `attenuationColor`, `clearcoat`, for gloss and translucency. Bake in the swatch textures.
 3. **Optional AI finishing:** pass the deterministic render through an image-edit model (Qwen-Image-Edit-2509 with depth/edge ControlNet, or FLUX/SDXL img2img at low denoise) **only for lighting, reflections and photographic feel**. Lock the zone masks and edge pixels.
 4. **Verification and correction:** sample each zone in the output, convert to CIELAB, compute ΔE_{2000} against the swatch reference, auto-correct (per-zone histogram/LUT transfer) or reject above a threshold (e.g., $\Delta E > 3$ for "presentation" and $\Delta E > 2$ for "sampling approval"; calibrate these thresholds with physical swatch tests). Always show the raw swatch next to the render.
 5. **Label renders** as "illustrative" vs. "color-verified" so nobody approves samples from an AI-finished image alone.

3. Function-by-function: which tool is right (as of late 2026)

Function	Right tool	Specific options	LLM needed?	Key risks
Sketch-to-render	Deterministic compositor/PBR	Three.js MeshPhysicalM	No (image model optional)	Color drift, invented

Function	Right tool	Specific options	LLM needed?	Key risks
	as truth; diffusion as optional finishing	aterial, Blender Cycles (headless, Python) for hero renders; Qwen-Image-Edit-2509 (Apache 2.0, native ControlNet depth/edge/sketch, multi-image input), FLUX.2 Klein (Apache 2.0), SDXL + ControlNet (lineart/canny/depth) + IP-Adapter; APIs: FLUX.1 Kontext [pro], Vizcom (for benchmarking)		geometry, licensing (FLUX.1 Kontext [dev] is under a non-commercial license)
Lamination zone handling	Classical vector geometry + rules; SAM for assistive selection	Paper.js/Konva boolean ops, Clipper2 offsets, OpenCV contours/fill; SAM 2 / SAM 3 (Meta, released Nov 19 2025; Meta's repo describes "a detector and a tracker that share a vision encoder" with "848M parameters"; text/exemplar/point prompts;	No	Line sketches are not natural images, so SAM may need click refinement; always store the result as vectors; check SAM License terms before production

Function	Right tool	Specific options	LLM needed?	Key risks
		weights under Meta's SAM License, not Apache 2.0) for click-to-select regions on bitmap sketches; Potrace/VTracer for bitmap → vector		
Color palette suggestion	Constrained optimization + small ML scorer; LLM for rationale	Color harmony in CIELAB/OKLCH ; candidate generation only from in-stock SKUs; scoring by trend-radar features + brand-DNA embeddings; LLM (GPT-class, Claude, Gemini, or open Qwen/Llama) to explain and to turn briefs into constraints	Helpful, not core	LLM inventing SKUs, so constrain it to tool calls against the stock DB
Supplier PDF/XLSX extraction	Document AI pipeline + VLM + human QA	Prefer XLSX/structured feeds (Mazzucchelli publishes its 2026 catalog as .xlsx); Docling (MIT, IBM-originated, TableFormer) or PyMuPDF for	Yes (VLM)	PDF color profiles and print/screen rendering make PDF swatch colors unreliable as truth, so use physical swatch scans with a color checker for "verified" colors;

Function	Right tool	Specific options	LLM needed?	Key risks
		layout/images; VLM (GPT-4o-class, Gemini, Qwen-VL) to map swatch image ↔ code ↔ name ↔ thickness; k-means/median color in Lab from cropped swatch		code mis-association
Brand DNA learning	Embeddings + parametric descriptors + style LoRA	Extract proportions from the signature library (lens width/height ratio, bridge, endpiece shape, temple profile) as structured features; CLIP/DINOv2/SigLIP embeddings in pgvector; LoRA on an Apache-licensed base for style finishing; Vizcom-style "palettes" as a benchmark	Partly (LLM to codify rules from designer interviews)	Small datasets mean overfitting; IP of training images; LoRA licensing follows the base model
Trend-to-design suggestion	Analytics/ML + LLM agent	Trend radar features (colors/materials /shapes frequency, growth) →	Yes (reasoning, briefs)	Spurious correlations; self-reported vendor accuracy

Function	Right tool	Specific options	LLM needed?	Key risks
		mapped to SKU attributes; optional data from Heuritech/WGS N-type providers or Mazzucchelli/Peclers trend books		
DFM guardrails	Deterministic rules engine + geometry checks	Shapely/Clipper 2 or OpenCascade (CadQuery) for width/offset checks; JSON-defined rules (json-rules-engine, or plain Python/pydantic) ; hinge library with min widths	No (LLM may explain violations)	False "pass" if rules are incomplete; rules must be co-signed with the manufacturer
Autonomous collection generation	LLM agent orchestrating deterministic tools	LangGraph / Temporal workflows; tools: stock query, palette generator, layer-stack builder, DFM check, renderer, ΔE verifier; outputs a tech pack (SKU list, stack, BOM, renders)	Yes	Agent loops, cost; must never bypass DFM/stock gates

4. DFM guardrails — concrete starting values (to be validated with your manufacturer)

Public sources give **no standardized minimum** for acetate around hinges, hidden-hinge insertion depth or endpiece thickness, so the rules engine must be parameterized per factory. Usable starting values:

- **Sheet thickness:** Mazzucchelli's own processing page lists block sheets at 3.0, 4.0, 6.0 and 8.0 mm and notes that with a 6 mm sheet, "it is common practice to glue rectangular blocks" to obtain nose pads and lugs, while with 8 mm sheets they are "obtained directly from the sheet by subtraction." It recommends planing "maximum 0.4mm for each passage." UK maker Banton Frameworks says fronts are "generally a 6/8mm thick section acetate" and temples "a thinner 4/6mm thick acetate." Laminated SKUs come in specific thicknesses (e.g., 906303 in 4.0/7.0 mm).
- **Sheet vs. finished thickness:** an 8 mm sheet can end up "6.5–7.5mm depending on bevel + polish" (Eyewearbeyond, an OEM blog). Raw sheets can vary "from 7mm to 10mm in the same sheet" before facing (Banton). For laminations, this means **the outer layers lose material to planing and polishing, which changes visible stripe proportions**. The rules engine must model the post-finish stack, not the nominal one.
- **Hinge anchoring:** hinge maker OBE publishes a "minimum width with plastic" per hinge, e.g., 2.8 mm for a 2.5 mm super-slim hinge, 6.0 mm for a 4.0 mm rivet-style wire hinge, 7.0 mm for a 5.0 mm hinge and 10.0 mm for an 8.0 mm hinge. In practice the acetate must be roughly 0.3–2 mm wider than the hinge. Rivets typically measure 0.5–2.0 mm in diameter (ZOYE catalogue); hidden heat-insert hinges come in 3–8 mm widths.
- **Lens groove:** about "1.5mm deep at an angle 120°" (Banton); one manufacturer on the OptiBoard forum cuts 2.5 mm, ending at 2.2 mm after tumbling. Another states "there is no standard dimension of the groove depth."
- **Tolerances:** ISO 12870 sets finished-frame tolerances of ± 0.5 mm for horizontal boxed lens size and distance between lenses and ± 2.0 mm for side length (2004/2016 editions; the 2024 edition is current). Credible acetate-specific CNC tolerances are not published; claims like " ± 0.005 mm" from OEM marketing are implausible, so use about ± 0.1 mm as a working design margin.
- **Rule examples to encode:** (a) endpiece width $\geq$ hinge "min width with plastic" for the selected hinge SKU; (b) any lamination layer that the rivet/hinge pocket would cut through must sit in a bonded zone of adequate thickness, with no hinge pocket straddling a glue line unless the factory approves; (c) total post-finish thickness $\geq$ groove depth + minimum wall on both sides; (d) temple stack thickness $\geq$ core-wire hinge width + wall; (e) the chosen SKU exists in the needed thickness (front vs. temple) with available sheet size/MOQ.

Details: Recommended Architecture

Data model (the heart of the system)

- **Material SKU:** supplier, code, name, collection, type (monocolor/havana/multilayer/laminated/gradient), available thicknesses, sheet sizes, MOQ, temple-suitability flag (Mazzucchelli explicitly flags some SKUs as "Thickness not made for Temples"), swatch images (front-lit, back-lit, edge), measured Lab, translucency class, sustainability variant (e.g., M49/Renew), embedding vector.
- **Layer stack:** an ordered list of {SKU, nominal thickness, finished thickness}, plus the bonding method and supplier pre-laminated SKU vs. custom-laminated.
- **Frame design:** vector front + temples (SVG paths), zones (closed paths → stack assignment), hinge SKU, rivet layout, lens tint, finish (gloss/matte).
- **Brand DNA:** signature models with parametric descriptors, assembly/finishing codes, shape-material compatibility rules, image embeddings.

Phase 3 stacks

Layer	Pragmatic MVP	Advanced	Trade-off
Frontend editor	React/Next.js + Konva (layers, hit-testing, transforms) or Paper.js (best vector boolean ops) for zone drawing; SVG import	Add Three.js/React-Three-Fiber 3D viewer with extruded geometry; WebGPU	Paper.js is stronger for path booleans; Konva is better for interactive layers. Fabric.js also works but has weaker booleans
Sketch ingestion	Potrace/VTracer vectorization; OpenCV cleanup; manual zone tracing	SAM 2/SAM 3 click-to-region; auto closed-region detection	Hand-drawn gaps break auto-fill, so keep manual tracing as the fallback
Rendering	2.5D compositor (Python/Pillow/Open CV or a WebGL shader): swatch texture per zone + procedural lamination stripes + normal/gloss maps from the extruded outline	Blender Cycles headless for hero renders; Three.js MeshPhysicalMaterial live; optional Qwen-Image-Edit/SD XL-ControlNet finishing via ComfyUI	Deterministic = exact but less "photographic"; diffusion = beautiful but must be verified

Layer	Pragmatic MVP	Advanced	Trade-off
Catalog ingestion	XLSX import + Docling/PyMuPDF + VLM mapping + human review UI	Scheduled supplier syncs; spectrophotometer/color-checker scans	Human QA is non-negotiable for "verified" colors
Backend	FastAPI (Python) + PostgreSQL + pgvector + S3-compatible storage; Celery/RQ jobs	Temporal or Prefect for long workflows; dedicated vector DB (Qdrant) only if scale demands	pgvector keeps everything in one database at MVP scale
GPU inference	Serverless GPU (Modal, Replicate, RunPod, fal) for occasional finishing	Dedicated L40S/H100 pool, TensorRT, model caching	Serverless costs per call; dedicated costs idle time
DFM engine	Python rules (pydantic + Shapely) running on every save	OpenCascade/CadQuery 3D checks; CNC toolpath simulation; factory-specific rule packs	Must be versioned and co-owned by the manufacturer
MLOps	Weights & Biases or MLflow; prompt/version registry; render audit log	Evaluation harness (ΔE , edge IoU vs. ground truth, designer ratings)	Without evals, quality regressions go unseen

Phase 4 stacks

- **Agent layer:** an LLM (API: GPT-class/Claude/Gemini; or self-hosted Qwen/Llama for confidentiality) orchestrated with LangGraph, calling *only* typed tools: `query_stock`, `propose_palette`, `build_stack`, `check_dfm`, `render`, `verify_color`, `score_trend`, `score_brand_fit`.
- **Scoring models:** gradient-boosted or small neural scorers trained on (design features $\times$ trend features $\times$ designer accept/reject feedback); brand-fit by embedding distance to signature models + rule compliance.
- **Generation loop:** brief $\rightarrow$ constraint set $\rightarrow$ N candidate stacks/palettes from stock $\rightarrow$ DFM filter $\rightarrow$ deterministic render $\rightarrow$ optional AI finishing $\rightarrow$ ΔE verification $\rightarrow$ ranked board with rationale $\rightarrow$ designer selection $\rightarrow$ tech pack export (SKU list, stack, hinge, BOM, renders) for sampling.

- **Human-in-the-loop by design:** "autonomous" should mean autonomous drafting with mandatory designer approval; every accept/reject becomes training signal.

Recommendations

1. **Pick the deterministic-first architecture now.** Make the layer-stack data model and procedural lamination renderer the MVP's core; treat diffusion as a toggleable enhancement.
2. **Start catalog ingestion with Mazzucchelli's XLSX** (2026 catalog) and its product pages (thickness, MOQ, laminated/multilayer classes). Then add PDF suppliers via Docling + VLM with a review queue. Buy physical sample cases (Mazzucchelli sells Creative/Sample Cases) and scan them under controlled light with a color checker to create "verified" Lab references.
3. **Run a 4-week rendering bake-off** on about 10 real Noe Noah sketches: (a) 2.5D compositor, (b) Three.js PBR extrusion, (c) compositor + Qwen-Image-Edit-2509 finishing, (d) Vizcom as an external benchmark. Measure per-zone ΔE_{2000} , lamination-edge accuracy against a physical sample, and designer preference.
4. **Co-write the DFM rule pack with your factory:** hinge SKUs and their minimum widths (OBE-style data), groove depth, planing allowance, minimum layer thickness at hinge pockets, temple core-wire constraints. Version it.
5. **License hygiene:** use Apache-2.0 bases (Qwen-Image-Edit, FLUX.2 Klein) for self-hosting or paid APIs with commercial output rights. Avoid FLUX.1 [dev]/Kontext [dev] weights in production without a BFL commercial license. Train brand LoRAs only on images Noe Noah owns.
6. **Explore partnership vs. build with VARAi** for the downstream CAD→3D→try-on leg; your defensible core is sketch-zone lamination, stock binding, DFM and brand DNA.
7. **Phase 4 only after Phase 3 data exists:** log every designer decision from day one; that dataset is what makes brand-DNA and trend scoring work.

Caveats

- Several competitor claims (VARAi "1:1-accurate," Vizcom "brand-accurate colors," Heuritech "92% accuracy") are vendor marketing and have not been independently verified.
- Color-control papers (NumColor, ColorWave, Paint-Anything) are recent research; Paint-Anything was seen only via a secondary blog summary. None prove supplier-level color fidelity on translucent acetate.
- DFM numbers come mostly from maker/OEM blogs and parts catalogues, not standards; ISO 12870's design annex was not publicly readable. Treat all values as starting points for factory validation.
- PDF swatch charts are print/screen approximations; only physical swatches measured under controlled conditions should be marked "color-verified."
- The model landscape changes monthly; re-check licenses and versions (e.g., newer Qwen-Image-Edit or FLUX.2 releases) at build time.

